

Data Engineer Interview Questions (2 to 5 Years of Experience)

1. What are the key responsibilities of a Data Engineer?
2. Explain the difference between OLTP and OLAP systems.
3. What are ETL and ELT? When would you use one over the other?
4. What are the common challenges in data engineering?
5. Explain the concept of data partitioning in a data warehouse.
6. How do you optimize a slow SQL query?
7. What is indexing in databases? How does it improve performance?
8. Explain different types of joins in SQL with examples.
9. What is denormalization, and when should you use it?
10. How do you handle duplicate records in SQL?
11. What is Apache Spark, and how does it compare with Hadoop MapReduce?
12. Explain the difference between Spark RDD, DataFrame, and Dataset.
13. What is the difference between `cache()` and `persist()` in Spark?
14. Explain Spark's shuffle operation and how to optimize it.
15. What are the different types of transformations in Spark?
16. Write a PySpark code to remove duplicates from a DataFrame.
17. Write a PySpark code to calculate the moving average of a column.
18. Write a Scala program to count the number of words in a text file using Spark.
19. Given a PySpark DataFrame, write a function to fill missing values with the column mean.
20. Write a PySpark program to group data by a column and calculate the sum of another column.
21. What is the difference between partitioning and bucketing in Hive?
22. How does Hive store data internally?
23. What are different file formats used in Hive?
24. Explain dynamic partitioning in Hive with an example.
25. How does indexing work in Hive?

26. What is Apache Airflow, and how is it used in data pipelines?
27. Explain how to schedule a pipeline in Airflow.
28. What are DAGs in Airflow?
29. How would you handle a failed task in an Airflow DAG?
30. What is backfilling in Apache Airflow?
31. What is the difference between AWS S3 and HDFS?
32. How does AWS Glue work, and when would you use it?
33. Explain how AWS Lambda can be used in data processing.
34. What is the difference between Redshift and Athena?
35. How do you optimize an S3 data lake for faster querying?
36. What is Apache Kafka, and how does it work?
37. Explain the difference between Kafka Producer and Consumer.
38. How do you ensure message ordering in Kafka?
39. What are Kafka topics and partitions?
40. How does Kafka handle fault tolerance?
41. What are some common Spark performance optimization techniques?
42. How do you optimize data storage in a data lake?
43. Explain the role of file formats like Parquet and ORC in performance tuning.
44. What is vectorization in Pandas and Spark, and how does it help in performance?
45. What are best practices for optimizing SQL queries on large datasets?